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Visual odometry in local underwater navigation problems

I.M. Azhmukhamedov, P.I. Tamkov , N.D. Svishchev , A.V. Rybakov
Astrakhan State University, Tatisheva 20a, Astrakhan, Russia

E-mail: svishchev.nikolai@gmail.com

Abstract. The work processes of the ORB-SLAM algorithm are presented. The results of experimental studies on temporal comparisons of the operation of the algorithm with different parameters and cameras are presented. The necessity of forming a visual odometry (VO) system as a local navigation of remote-controlled and autonomous underwater robots has been substantiated. The two most suitable odometry methods in the underwater environment are described, such as their advantages and disadvantages. The work processes of the ORB-SLAM algorithm are presented. The results of experimental studies on temporal comparisons of the operation of the algorithm with different parameters and cameras are presented. The procedure for preparing video data is described: processing a video stream, adjusting camera parameters for calibration. The experiments represent the testing of the ORB-SLAM3 algorithm on a sample of video filmed as part of the ecological monitoring of the Caspian shelf in 2020.

1. Introduction

In the field of navigation with the use of computer vision, many methods have been developed for constructing dense three-dimensional maps based on depth cameras. The SLAM (simultaneous localization and mapping) visual system algorithms have gained great popularity in the last decade, the key factor being the system's ability to obtain metric, brightness, color, texture information about the environment and create accurate and realistic 3D maps. However, environmental mapping methods can be inaccurate due to inadequate camera projection models.

Descriptors form the core of many machine vision algorithms with real-time requirements such as object detection, visual odometry, or SLAM tasks. It is usually assumed that the resulting images, and therefore the areas highlighted around the key points, come from a perspective projection that ignores image distortion, or projections inherent in wide-angle lenses. Deviations from the ideal perspective projection are often corrected using standard compensation equations. The latter, however, cause artifacts if the camera's field of view is enlarged. In addition, many applications (eg. monocular odometry) require keypoints only on an undistorted projection.

2. Literature review

In June 2020, one of the extreme modifications of the visual odometry algorithm was published, called: ORB-SLAM 3 [1]. This method has three main parallel streams: a tracking stream, a local mapping stream, and a map compensation and merging stream. Most modern works do not take into account the features of optical distortions of lenses of used wide-angle cameras, this is evidenced by a number of works on the study of some of the most popular methods: ORB-SLAM, ORB-SLAM 2, ORB-SLAM 3 [1-3].

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2.1. Camera models in SLAM algorithms

Since the first publication of the ORB-SLAM algorithm in 2015 [1] and at its extreme configuration, with the addition of data from the inertial navigation module, in 2021 [3], the authors of the works have made a significant breakthrough in the investigated problems of SLAM. This model projects hemispherical lens distortions onto a flat projection of the image, thereby generating artifacts in the frames caused by this transformation [4]. Distortion compensation methods in modern research suggest the presence of robust methods that eliminate interference when using computer vision algorithms [5]. This approach shows good results in the speed and reliability of the entire system by reducing computations and artefacts in the output image.

2.2. Current state of Visual Odometry research

Significant research interest is aimed at ensuring the autonomy of mobile systems using computer vision, namely, at self-determination of location and assessment of movement, as evidenced by a number of published works [6 - 12]. According to the tendency of publications, automatic retention of the position relative to the visually tracked object is of less interest to scientists in comparison with navigation and solving SLAM problems in conditions of an optically inhomogeneous medium [7, 9, 12]. The works related to the classification and relative orientation of autonomous mobile complexes were mainly aimed at visual tracking of objects of interest by means of digital and analog cameras [6, 11]. Basically, the authors of the works operated both with a three-dimensional cloud of points obtained using a stereo pair of cameras and using monocular vision using additional navigation aids, such as inertial navigation modules, echo sounders, laser pointers and other navigation systems [13 - 20]. The analysis of the presented works revealed the lack of results on the application of the algorithms under study in conditions of dynamic structural noise. In the overwhelming number of works, numerous methods and algorithms for recovering from interference of underwater images (color correction, contrast, separation of object boundaries, restoration of transparency and image illumination) were considered [21 - 28]. In works devoted to the restoration of the quality of underwater images, both artificial neural networks [21, 28] and classical methods of image preprocessing, as well as their combination, were used.

2.3. The future of underwater visual odometry algorithms

Due to the limited underwater visibility range, capturing images at close range is often the only solution for obtaining large visible areas of a textured surface. This limitation has motivated research to improve visual odometry techniques over the past several years. In the works mentioned above, an installation was proposed for creating two-dimensional and three-dimensional maps by obtaining images and fixing landmarks in them, along which the map is glued or the distance travelled is calculated. A common problem when mosaicking SLAM problems arises from the presence of distortion and dynamic structural noise in the environment. Strategies for dealing with such occlusions are being discussed. Another difficulty arises from the proliferation of alignment errors.

3. Investigation of methods for constructing a camera model and distortion

All SLAM algorithms require accurate camera calibration. First of all, the internal parameters of the camera / lens system are extracted, which consists of the focal length matrix and the optical centre of the camera (f_x, f_y, c_x, c_y), the rotation matrix R and displacement t, and 5 parameters of the distortion function $(k_1, k_2, k_3,\ p_1,\ p_2)$. Where k_1, k_2, k_3 are the radial distortion coefficients. Radial distortion - image distortion as a result of the non-ideal parabolic shape of the lens and the optical properties of the medium. Distortions caused by the radial distortion of the lens are equal to 0 at the optical center of the sensor and increase towards the edges, while the consideration of the radial distortion of the medium will be described below. As a rule, radial distortion contributes the most to image distortion. $p_1,\ p_2$ - coefficients of tangential distortion. Tangential distortion - image distortion caused by errors in the installation of the lens parallel to the image plane [29].

3.1. Projection of a 3D point onto a 2D image surface

In order to find the projection of a point in space onto the image plane, you first need to translate the point (X_w, Y_w, Z_w) from the world coordinate system to the camera coordinate system, taking into account external parameters (rotation matrix R and displacement t) (Figure 1).

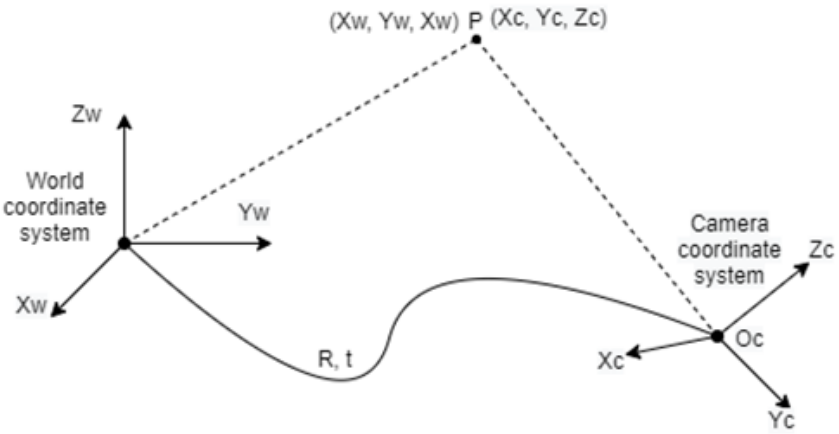

Figure 1. The world coordinate system and the camera coordinate system are linked by rotation and movement. These six parameters (3 for rotation and 3 for movement) are called external camera parameters.

Next, using the internal parameters of the camera, we project the point onto the image plane. The equations that relate a 3D point (X_w, Y_w, Z_w) in world coordinates with its projection in image coordinates (u, v) are written below:

$$\begin{bmatrix} u' \\ v' \\ w' \end{bmatrix} = P \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}, \tag{1}$$

where the coordinates of the image are:

$$u = \frac{u'}{w'}, v = \frac{v'}{w'} \tag{2}$$

where P is a 3 × 4 projection matrix consisting of two parts: the inner matrix K, which contains the internal parameters, and the outer matrix $(R|t)$, which is a combination of the 3 × 3 rotation matrix R and the 3 × 1 transformation vector t.

$$P = K[R|t], \tag{3}$$

where the internal matrix K, is equal:

$$K = \begin{bmatrix} f_x & \gamma & c_x \\ 0 & f_x & c_y \\ 0 & 0 & 1 \end{bmatrix}, \tag{4}$$

f_x, f_y - focal lengths x and y in pixels, as a rule, have the same value. c_x, c_y - coordinates x and y of the optical center in the image plane (usually located in the center of the image). Using the center of the image is usually a good enough approximation [30], γ - the misalignment between the axes. Usually equal to 1, in most calibration packages such as *OpenCV* this factor is equal 0.

Thus, if the camera image is zoomed in or out due to certain factors, all of these parameters $(f_x,\ f_y,\ c_x,\ c_y)$ will be scaled (multiplied or divided) to the same scale. The matrix of internal parameters does not depend on the type of scene, and after calculation it can be reused (provided that the focal length is fixed).

3.2. Lens distortion model

To find the parameters of the distortion function, researchers generally resort to the classical Brown-Conrady model, which describes the transition from an undistorted image to an image with distortion by the following transformations of coordinates (x_u, y_u) of a point on an undistorted image:

$$\begin{cases} x_d = x_u(1 + k_1 r^2 + k_2 r^4 + \cdots) + (p_2(r^2 + 2x_u^2) + 2p_1 x_u y_u) \\ \qquad (1 + p_3 r^2 + p_4 r^4 + \cdots); \\ y_d = y_u(1 + k_1 r^2 + k_2 r^4 + \cdots) + (p_1(r^2 + 2y_u^2) + 2p_2 x_u y_u) \\ \qquad (1 + p_3 r^2 + p_4 r^4 + \cdots), \end{cases}, \tag{5}$$

where (x_d, y_d) - coordinates of the point (x_u, y_u) on the distorted image; $r = ((x_u - c_x)^2 + (y_u - c_y)^2)^{\frac{1}{2}}$ - distance from the optical center (c_x, c_y) to point (x_u, y_u); k_i - radial distortion coefficients; p_i - tangential distortion coefficients.

In practical problems, the coefficients of tangential distortion are considered up to the coefficient p_2 inclusive, and the radial - up to coefficient k_3 or k_6 inclusive. Thus, the radial component in the tangential distortion is almost always omitted.

The *OpenCV* library (which is the world standard for computer processing of images and video in the field of free software) the following distortion model is used (6).

It should be especially noted that in *OpenCV* the coefficients of the radial distortions $k_{4\ldots6}$ are not used as coefficients for higher degrees of the polynomial (as in the classical description (5)), and as the coefficients of a similar polynomial in the denominator. This type of radial distortion function reduces the probability of exceeding the maximum the possible value of the variable when calculating the twelfth power numbers r:

$$\begin{cases} x' = \Delta_r x + \Delta x_t; \\ y' = \Delta_r y + \Delta y_t; \end{cases} \tag{6}$$

where $\Delta_r = \frac{1 + k_1 r^2 + k_2 r^4 + k_3 r^6}{1 + k_4 r^2 + k_5 r^4 + k_6 r^6}$ - coordinate displacement caused by radial distortion; $\Delta x_t = p_2(r^2 + 2x^2) + 2p_1 xy$ - offset of the x coordinate caused by tangential distortion; $\Delta y_t = p_1(r^2 + 2y^2) + 2p_2 xy$ - displacement of the y coordinate caused by tangential distortion; $r = (x^2 + y^2)^{\frac{1}{2}}$.

Thus, for modeling [31] (for the purpose of subsequent compensation) the distortion requires the following input data (7). It can get using the *calibrateCamera* function of the *OpenCV* library by

images of the calibration template obtained from the calibrated camera:

$$A = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}; D = \begin{pmatrix} k_1 \\ k_2 \\ p_1 \\ p_2 \\ k_3 \\ \ldots \end{pmatrix}, \tag{7}$$

where A - camera matrix (used to normalize pixel coordinates); D - vector of distortion coefficients; (f_x, f_y) - reduced lens focal lengths along two axes.

The Brown – Conrady model allows obtaining distorted coordinates of a point from undistorted ones, and this is enough to obtain an image with compensated distortion from a distorted image.

To eliminate distortion in the image, you can simulate the distortion on a uniform rectangular point map, associate it with the distorted image, and then transform the image so that the point map becomes uniformly rectangular again.

However, the problem often arises to obtain the coordinates of a point in an image with a compensated distortion by the coordinates of a point in a distorted image without transforming the entire image (since this operation has a high computational complexity).

Ideally, it is necessary to find a solution to the polynomial, but its degree is too high, which makes an analytical solution impossible. However, a numerical solution can be found using the iterative algorithm described above. With each iteration, the values (x, y) approach to (x_u, y_u).

4. Experiments

In the visual SLAM system, one of the main problems is the selection of key points in the image that can be used as reliable reference points. These points must be invariant to rotation, movement, scaling, projection distortion, and uneven lighting. In practice, these requirements are not always met, and the problem arises of finding a match for the same key points on different images [7 - 10].

4.1. Investigation of camera calibration in underwater visual odometry

To investigate the effect of the camera model on the performance of visual odometry underwater, a SLAM algorithm called ORB-SLAM3 was used. This algorithm was chosen based on the following considerations:

• Easy to configure internal camera parameters;
• Visual influence of camera parameters on the work of visual odometry;
• Easy to customize your database;
• Options for running the algorithm in both online and offline modes;
• Various variations of the cameras used, monocular camera, stereo pair, joint work with inertial navigation system to compensate for camera turns in three corners.

The video from the GoPro 4 camera was taken as a basis, the movement of the camera was carried out by means of a remote-controlled unmanned underwater vehicle moving in the water column with a direct line of sight to the relief of the bottom of the Caspian Sea (Figure 2). The robot moved along a relatively straight trajectory, with a periodic change in the distance to the ground ± 0.5 m. The ground is a mixture of a bat and a whole shell with small bas-reliefs 10-20 cm. The transparency of the water in the thickness at the time of shooting was 2 meters, the transparency was assessed using a disk Secchi.

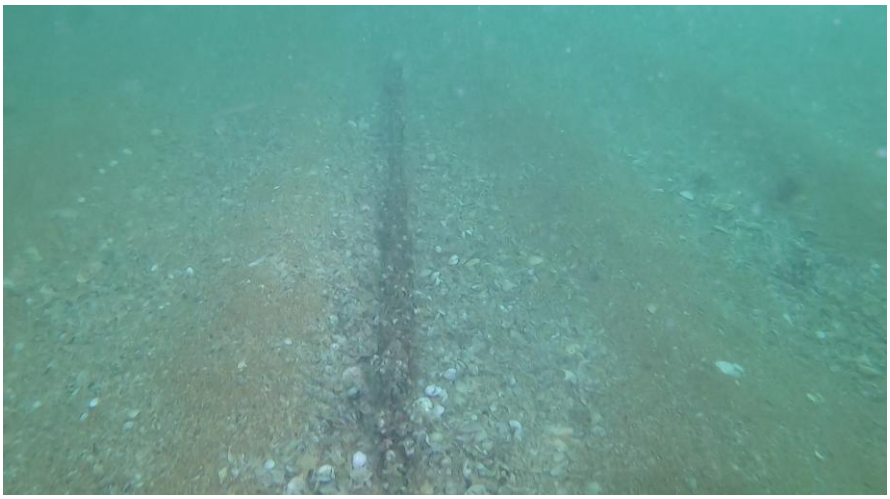

Figure 2. A frame of underwater video taken for testing the ORB-SLAM3 algorithm.

Given the fact that the video was shot in a high density environment, the distortion coefficients should be different from those obtained during the calibration of the GoPro camera using the *OpenCV* library.

This paper analyzes two methods for eliminating distortion in underwater images. The first, numerical, is to calibrate the camera in a standard model directly under water. The second, analytical one, consists in correcting underwater radial distortion using parameters obtained to eliminate

distortions above water, and applying a correction that takes into account the refractive index of the medium different from air.
The numerical method assumes calibration directly under water, while the second assumes the presence of only the value of the density of the medium, which is the optimal method within the framework of our problem, but within the framework of constructing a trajectory based on visual odometry, this method will lead to a rapid accumulation of errors.

After obtaining the calibration values of the GoPro 4 camera using the *OpenCV* library and entering these data into the camera configuration file of the ORB-SLAM3 algorithm (Table 1).

Table 1. GoPro Camera Model Coefficients 4.

Parameter	Parameter value
Lens model	Pinhole
f_x	458.654
f_y	457.296
c_x	367.215
c_y	248.375
k_1	-0.283408
k_2	0.073959
p_1	0.000193
p_2	1.761871

The data presented above does not take into account the radial distortion of the medium, so you can clearly see the effect of this distortion in the field (Figure 3).

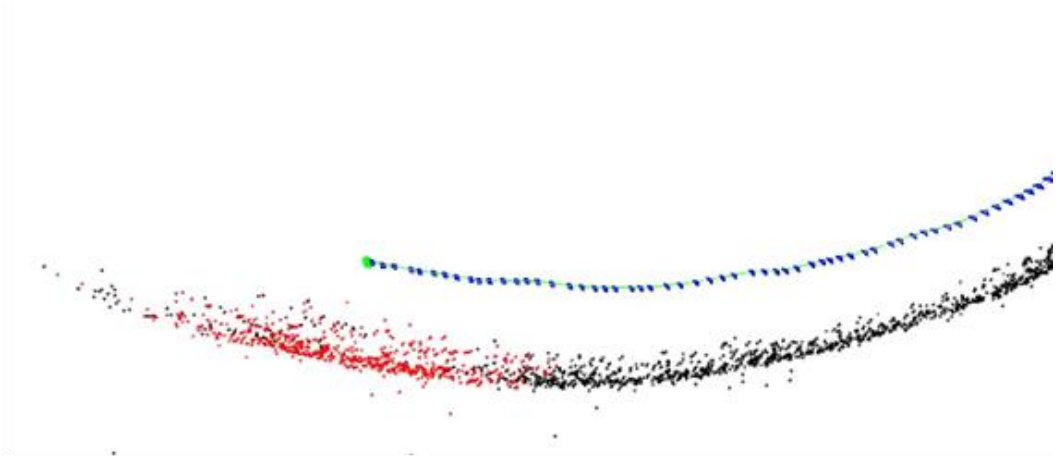

Figure 3. Black dots are the key points of the frame selected for constructing the trajectory, red dots are the key points of the current frame, along which the trajectory is calculated, and the dashed line is the calculated trajectory of movement along the frames of the video stream.

Given the fact that the underwater robot is moving straight in the video, nevertheless, when building a trajectory using visual odometry, the movement curve has a strong curvature with a bulge downward, which tells us that we are moving inside the pit, which is not true. This effect can be explained by the fact that although we have the true calibration data of the camera lens, we nevertheless did not take

into account the radial distortion of the environment, which gives such an error in the analysed SLAM algorithm.
With a known refractive index of water, the correction of the distortion that occurs when the camera is immersed is described by transforming the coordinates of the image in the plane of the pinhole camera screen:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \frac{1}{(1-(n^2-1)(x^2-y^2))^{\frac{1}{2}}} \begin{bmatrix} x \\ y \end{bmatrix}, \tag{8}$$

where $(x\ y)^T$ and $(x'\ y')^T$ are the coordinates of the object on the screen of the pinhole camera when shooting over water and under water, respectively, and n is the refractive index of water.

The calculated coefficients of the radial distortion of the medium, together with the lens distortion, give new values $k_1 = 0.262383$ and $k_2 = -0.9531$, respectively. When the algorithm is restarted, taking into account the coefficients, we will obtain the following trajectory (figure 4).

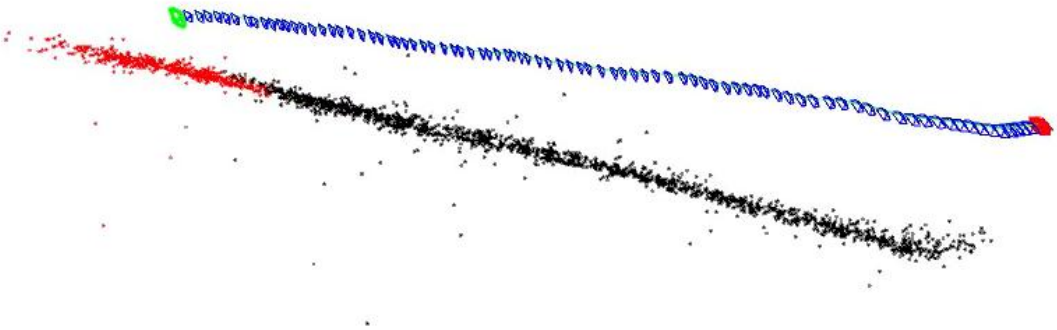

Figure 4. Black dots are the key points of the frame selected for constructing the trajectory, red dots are the key points of the current frame, along which the trajectory is calculated, and the dashed line is the calculated trajectory of movement along the frames of the video stream.

However, the accuracy of the correction calculated in this way, as well as its resistance to the error in determining the refractive index, require more detailed study. Including in the context of the most significant case from a practical point of view - underwater photography.

5. Conclusions and further research prospects

In this work, a study was carried out of the influence of the optical properties of the aquatic environment on the visual odometry algorithm of an underwater robot. This study proves the lack of robustness of the classical methods of camera calibration and accounting for environmental distortion in the operation of underwater visual odometry. Further developments involve the introduction of a generative adversarial neural network (GAN) into existing algorithms, the rejection of the camera distortion transformation model in favor of using a distortion compensation model, the rejection of the current least squares methods that are not resistant to spurious outliers when constructing graph adjacency matrices in SLAM problems in favor of improving the positioning algorithm based on graphs SLAM, this approach will allow developing robustness to outliers in SLAM problems. To

achieve this goal, it is proposed to conduct the following series of complex studies on the selected methods.

The first study is aimed at general improvement of 3D mapping algorithms in an optically heterogeneous and noisy environment, with special attention to improvement methods based on graph based SLAM graphs. The second approach is to develop methods for compensating for distortion that occurs when using wide-angle lenses of a stereo pair of digital cameras and the features of an external optically inhomogeneous medium. The main method that can be relied on in this approach is the existing high-speed detector FAST (Features from Accelerated Segment Test), which, within the framework of our study, needs to be modified for wide-angle lenses. In addition, it is necessary to develop a descriptor based on the existing ORB (Oriented FAST and rotated BRIEF), taking into account distortion. Typically a fisheye camera is used to capture wide-angle and full-screen images, but such wide images are not invariant to camera rotation, that is, the distortion of the scene captured by the camera depends on the image area. On the other hand, such distortions are static, and they can be taken into account in computer vision algorithms without attempting to project onto a flat projection, preserving the integrity of the data obtained and increasing the speed of the algorithm.

The third stage is the improvement of algorithms for filtering dynamic structural noise arising due to the presence of suspended particles in the medium that reduce the computer vision field. The solution to the problem of removing image artifacts from a stereo camera is based on the use of a GAN. The traditional way of evaluating algorithms for image reconstruction and enhancement tasks is to measure distortion, the difference between true images and reconstructed images, using metrics such as mean square error (MSE), peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM). However, these methods are not able to solve the problem of filtering dynamic structural noise arising due to the presence of particles of different sizes in the medium. Structural interference of this type can be divided into two classes, the first class of interference consists of very small particles that are not detected by the SLAM algorithm as key points, but strongly narrow the field of view of the camera. The second type of structural interference is large dynamically moving particles that introduce errors in SLAM navigation by creating new feature points. To eliminate the first type of interference, it is proposed to use image recovery algorithms based on GAN, while one of the difficult tasks is to minimize the computational resources required to use this algorithm. The second type of interference leads to emissions during SLAM operation and can be compensated for by an adaptive controller that takes into account the expected direction of travel and the readings of the inertial navigation system.

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